Trekking Management Project Report

1. Student Details

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About Me: I am a dual degree undergraduate student currently pursuing a BS in Data Science. I am interested in backend web development and database-driven applications. Through this Placement Portal project, I gained practical experience in building a role-based web application using Flask, SQLAlchemy, and SQLite, along with designing database relationships and implementing dashboard-based management features.

2. Project Details

Project Title: Trekking Management Application

Problem Statement:

The Trekking Management Application is a web application developed to simplify trekking operations.The system connects three types of users: **Admin, Trekkers, and Staffs**.
It enables administrators to manage **treks**, **staff**, and **bookings**, **trek staf**f to handle assigned **treks**, and users to browse and **book treks**.
The system provides secure role-based access, efficient booking management, and organized trekking records.

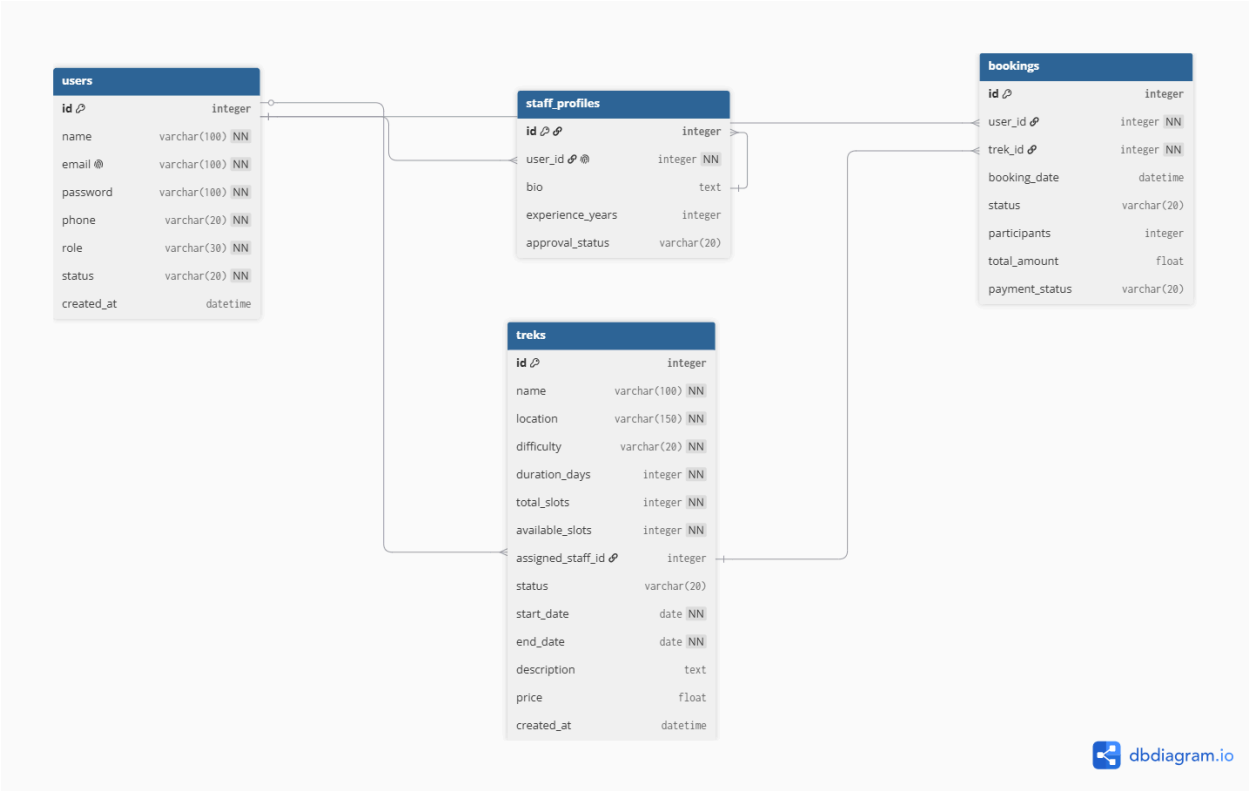
3. AI / LLM Declaration

I used ChatGPT (GPT-5) for assistance during the development of this project. The assistance included understanding Flask concepts, debugging suggestions, improving HTML/CSS layout.The extent of AI/LLM usage is approximately **35–40%**, mainly for guidance and code suggestions and debugging.

4. Technologies and Frameworks Used

Technology / Library	Purpose
Flask	Core backend web framework
SQLAlchemy	Object Relational Mapper(ORM) for database operations
Jinja2	Template engine for rendering dynamic HTML Pages
HTML	Structure of application web pages
CSS	Styling and layout of the application
SQLite	Lightweight relational database for storing application data
Flask Sessions	Used for authentication and role-based access control

5. Database Schema / ER Diagram



6. Functionalities

Authentication

- Users and trek staff can register and log in.
- Admin account is pre-created and manages the system.
- Role-based authentication for Admin, Trek Staff, and Users.

Admin Features

- Add, edit, and remove treks.
- Add, approve, and assign trek staff to treks.
- View and manage users, staff, treks, and bookings.
- Search users, staff, or treks.
- Blacklist users or staff if required.

Trek Staff Features

- View assigned treks.
- Update trek status and available slots.
- View registered participants for assigned treks.
- Mark treks as completed.

User Features

- Register and log in to the application.
- Browse, search, and filter available treks.
- Book and cancel trek bookings.
- View booking status and trekking history.

Additional Features

- Prevention of duplicate trek bookings.
- Prevent overbooking beyond available slots.
- Role-based access control.
- Booking history tracking.
- Automatic update of trek slots after booking or cancellation.

7. Video Presentation

Drive Link:

https://drive.google.com/file/d/1Ma_KZd0nMAmATIR5A7O3OjKl_Z95Tlsp/view?usp=drive_link